INCH-POUND
MIL-PRF-83513/20B
15 August 1997
SUPERSEDING
MIL-C-83513/20A
14 October 1993

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, RIGHT ANGLE, SOCKET CONTACTS, 3 ROW ,SOLDER TYPE, STANDARD PROFILE, 51 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-83513.

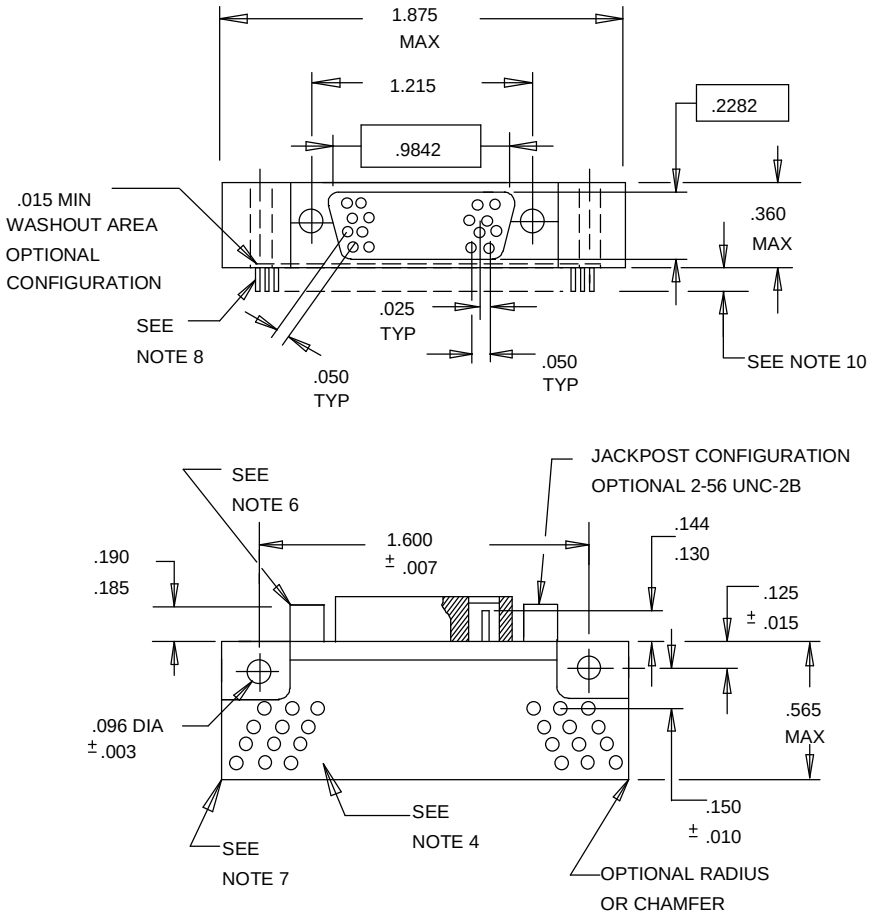
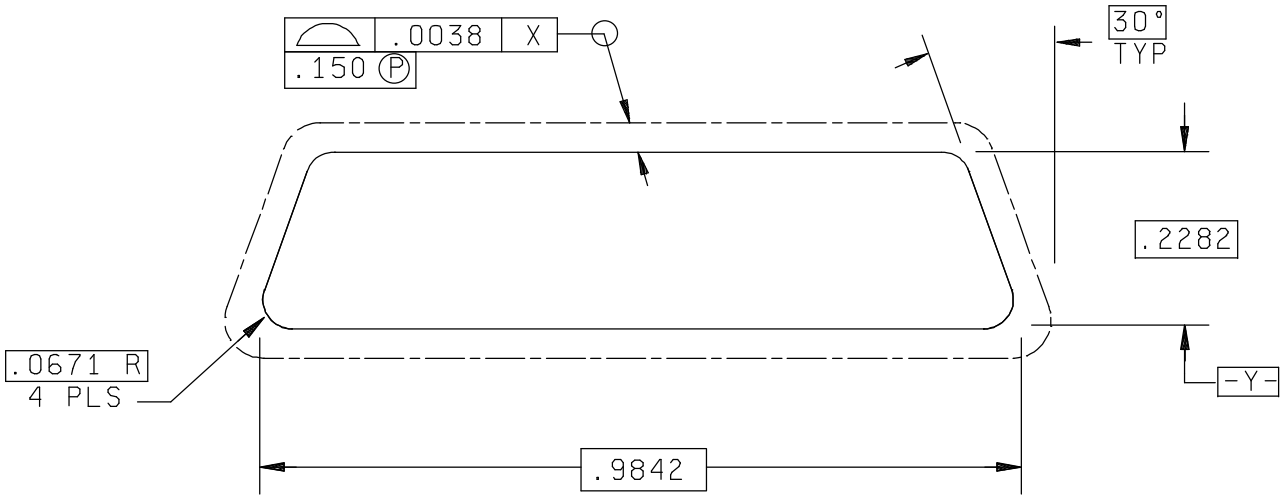


FIGURE 1. Connector, receptacle, .050 spacing.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.



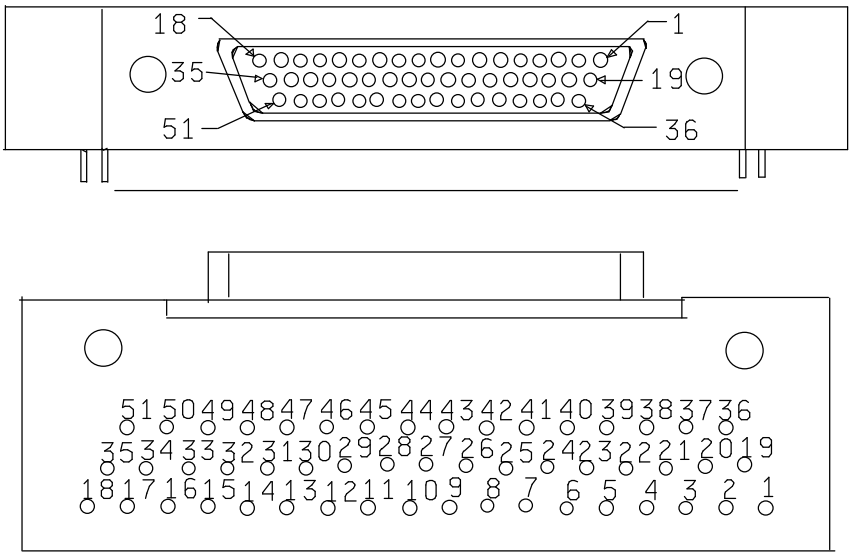
Inches	mm	Inches	mm	Inches	mm
.003	0.08	.0671	1.704	.360	9.14
.0038	0.097	.096	2.44	.565	14.35
.007	0.19	.125	3.18	.9838	24.999
.010	0.25	.150	3.81	1.215	30.36
.015	0.38	.185	4.70	1.600	40.64
.025	0.64	.190	4.83	1.875	47.63
.050	1.27	.2282	5.796		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-PRF-83513. Plastic molding shall conform to the requirements of Diallyl phathalate or Thermoplastic polyphenylene sulfide or Thermoplastic polyester, glass reinforced liquid crystalline.
5. Metal shell shall be of material in accordance with MIL-PRF-83513 for class M parts.
6. Jackpost (permanently attached), when specified: Corrosion resistant steel in accordance with ASTM A 484 and ASTM A 582, 300 series stainless steel, passivated in accordance with ASTM A 967.
7. Separately molded plastic body (if used) shall conform to the requirements of Diallyl phathalate or Thermoplastic polyphenylene sulfide or Thermoplastic polyester, glass reinforced liquid crystalline.
8. Wire termination sockets shall conform to QQ-W-343, no. 24 AWG copper.
9. Interfacial seals are included with each type receptacle connector.
10. Termination lengths available: .109 (2.77 mm), .140 (3.56 mm), or .172 (4.37 mm). The tolerance shall be ±.015 (0.38 mm) for all termination lengths.

FIGURE 1. Connector, receptacle, .050 spacing - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.



TERMINATION VIEW-RIGHT ANGLE
STANDARD PROFILE

NOTE:
1. Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement.

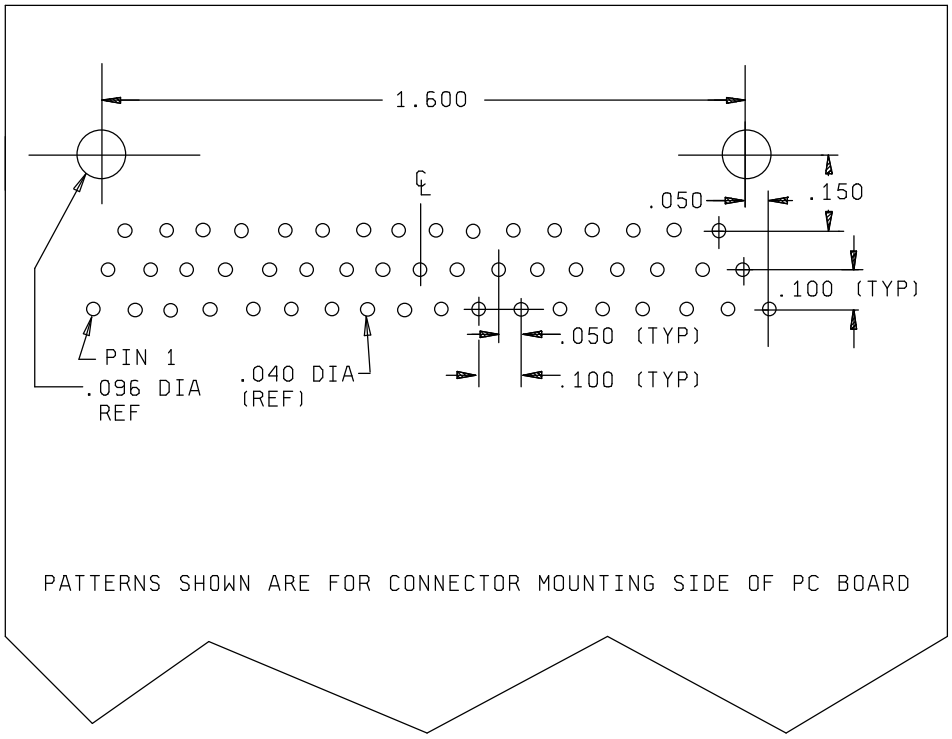


FIGURE 3. Layout arrangement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Current rating, maximum: 3 amperes per contact.

Materials:

Termination organization area: Potting fill material capable of passing the electrical and environmental requirements of MIL-PRF-83513.

Shell: The requirements for shell materials shall be in accordance with MIL-PRF-83513.

Plastic body or plastic molding: Shall conform to the requirements of Diallyl phathalate or Thermoplastic polyphenylene sulfide or Thermoplastic polyester, glass reinforced liquid crystalline.

Jackpost: Corrosion resistant steel in accordance with ASTM A 484 and ASTM A 582, 300 series stainless steel, passivated in accordance with ASTM A 967.

Wire termination pins: Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper.

Mating connector: Shall conform to MIL-PRF-83513/1 and MIL-PRF-83513/3.

Plating of termination leads: Solder dipping socket of termination leads will be accomplished in SN60 Pb40 or SN63 Pb37 in accordance with J-STD-006.

Part or Identification Number (PIN): The PIN shall consist of the letter M, the basic number of the specification sheet, a letter from the insert, a numerical code for the termination length, and a letter code for the shell finish and hardware column.

M83513/20-	G	01	C	P
Specification sheet number	Insert arrangements (see figure 2) G=51	Termination length 01=.109 02=.140 03=.172	Shell finish (Interface critical) C = cadmium N = electroless nickel (space application)	Hardware N = no jackpost P = jackpost attached

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA-NA

Review activities
Army - AT, MI
Navy - AS, CG, MC, SH
Air Force - 17, 99

Preparing activity:
DLA - CC

(Project: 5935-4087-19)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.